

Imperial College London

MSc EXAMINATION June 2023

*This paper is also taken for the relevant Examination for the Associateship*

## MRES - STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS

**For MLBD MRes Students**

June 5th, 2023: 10:00 to 12:00

*Answer all FOUR questions*

*Marks shown on this paper are indicative of those the Examiners anticipate assigning.*

### **General Instructions**

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

**You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.**

1. The Maxwell-Boltzmann distribution describes the distribution of the speed,  $v \in [0, +\infty)$ , of molecules in a gas at temperature  $T$ . Its probability density is given by,

$$f(v; T) = \left( \frac{m}{2\pi kT} \right)^{3/2} 4\pi v^2 e^{-\frac{mv^2}{2kT}},$$

where  $m$  is the molecular mass and  $k$  is a constant known as the Boltzmann constant. The speed  $v$  of any molecule will be a random variable  $v \sim f(v; T)$ .

- (i) Show that the first algebraic moment (the expectation of  $v$ ) of the Maxwell-Boltzmann distribution  $\mu_1$  is given by,

$$\mu_1 = E[v] = 2 \sqrt{\frac{2kT}{m\pi}}.$$

You may use the following identity,

$$\int_0^{+\infty} x^3 e^{-bx^2} dx = \frac{1}{2b^2}.$$

**ANSWER: UNSEEN:** The first algebraic moment is defined by,

$$E[v] = \int_0^{+\infty} f(v; T) v dv = \int_0^{+\infty} \left( \frac{m}{2\pi kT} \right)^{3/2} 4\pi v^3 e^{-\frac{mv^2}{2kT}} dv.$$

[1]. Noting that the integral is now in the form given in the identity, we set  $b = m/2kT$  [1] and using the identity we have that,

$$E[v] = \left( \frac{b}{\pi} \right)^{3/2} 4\pi \int_0^{+\infty} v^3 e^{-bv^2} dv = \left( \frac{b}{\pi} \right)^{3/2} \frac{4\pi}{2b^2} = 2 \sqrt{\frac{1}{\pi b}} = 2 \sqrt{\frac{2kT}{m\pi}}.$$

[1]. [3 marks]

- (ii) Imagine an experimental apparatus designed to determine the speed  $v$  of a molecule in a gas. This is possible at least in terms of measuring small intervals of speeds but don't worry about the details. Suppose the experiment observes a large number  $n$  of speeds  $\{v_1, v_2, \dots, v_n\}$ . These observations are independent and identically distributed random variables distributed as  $v_i \sim f(v; T)$ .

- (a) Show that the sample mean,  $\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$ , is an unbiased estimator for the first algebraic moment of the the Maxwell-Boltzmann distribution.

**ANSWER: UNSEEN:** The first algebraic moment is just the expectation value from the previous question so we set that equal to the sample mean to estimate it,

$$\hat{\mu}_1 = \bar{v} = \frac{1}{n} \sum_{i=1}^n v_i.$$

[This question continues on the next page ...]

[1] The definition of bias is the difference between the expectation value of the estimator and the true value.

$$\text{Bias}(\hat{\mu}_1) = E[\hat{\mu}_1] - \mu_1 = E\left[\frac{1}{n} \sum_{i=1}^n v_i\right] - \mu_1.$$

This can either be written as above or in words “if the estimator is unbiased, then the expectation value of the estimator is equal to the true value”. [1] By the linearity of the expectation operator, we have

$$E\left[\frac{1}{n} \sum_{i=1}^n v_i\right] = \frac{1}{n} \sum_{i=1}^n E[v_i],$$

[1] and since  $E[v_i] = \mu_1$ , we have that,

$$E\left[\frac{1}{n} \sum_{i=1}^n v_i\right] = \frac{1}{n} \sum_{i=1}^n \mu_1 = \mu_1 \frac{n}{n} = \mu_1,$$

since  $\mu_1$  is a constant. Plugging this into the equation for the bias above, we find that  $\text{Bias}(\hat{\mu}_1) = \mu_1 - \mu_1 = 0$  - i.e the estimator is unbiased [1]. [4 marks]

- (b) Using the method of moments, show that an estimator for the temperature of the gas  $\hat{T}$  based on the sample mean, is given by,

$$\hat{T} = \frac{\pi m \bar{v}^2}{8k}.$$

**ANSWER:** SEEN in Lectures (MoM): In this case, we need to equate the first algebraic moment in terms of  $T$  with the sample mean,

$$\mu_1 = \bar{v} \implies 2 \sqrt{\frac{2k\hat{T}}{m\pi}} = \bar{v}$$

the right-hand side of which can be re-arranged so that,

$$\hat{T} = \frac{\pi m \bar{v}^2}{8k}.$$

[1] [1 mark]

- (c) Is the estimator  $\hat{T}$  a biased or unbiased estimator of  $T$ ? Hint: Use the fact that,

$$\bar{v}^2 = (\bar{v} - \mu_1)^2 + 2\mu_1(\bar{v} - \mu_1) + \mu_1^2,$$

and assume that the second central moment (the variance) of the Maxwell-Boltzmann distribution,  $v_2 = E[(v - \mu_1)^2]$ , is non zero.

[This question continues on the next page ...]

**ANSWER: UNSEEN:** In order to calculate the bias we need to know the expectation value of  $\hat{T}$ . First, we determine,

$$E[\hat{T}] = E\left[\frac{\pi m \bar{v}^2}{8k}\right] = \frac{\pi m}{8k} E[\bar{v}^2].$$

[1] Focusing on the term inside the expectation value, we find,

$$E[\bar{v}^2] = E[(\bar{v} - \mu_1)^2 + 2\mu_1(\bar{v} - \mu_1) + \mu_1^2] = E[(\bar{v} - \mu_1)^2] + 2\mu_1 E[\bar{v} - \mu_1] + \mu_1^2$$

where we've used the expansion of  $\bar{v}^2$  given in the question [1] and the linearity of the expectation [1]. The first term is the variance of the sample mean,

$$E[(\bar{v} - \mu_1)^2] = V(\bar{v}) = V\left(\frac{1}{n} \sum_{i=1}^n v_i\right) = \frac{1}{n^2} V\left(\sum_{i=1}^n v_i\right),$$

where we used the fact that  $V(aX) = a^2 V(X)$  seen in lectures [1]. Next, we use the fact that the variance of a sum is the sum of the variances so that we have,

$$E[(\bar{v} - \mu_1)^2] = \frac{1}{n^2} \sum_{i=1}^n V(v_i) = \frac{n}{n^2} v_2 = \frac{v_2}{n},$$

where  $v_2$  is the variance of the Maxwell-Boltzmann distribution [1]. The next term is zero since,

$$E[\bar{v} - \mu_1] = E[\bar{v}] - \mu_1 = \frac{1}{n} \sum_{i=1}^n E[v_i] - \mu_1 = \mu_1 - \mu_1 = 0.$$

[1] We then have,

$$E[\bar{v}^2] = \frac{v_2}{n} + \mu_1^2.$$

Inserting this into the expectation value of the estimator, we get

$$E[\hat{T}] = \frac{\pi m}{8k} \left( \frac{v_2}{n} + 4 \cdot 2 \frac{kT}{\pi m} \right) = \frac{\pi m}{8k} \frac{v_2}{n} + T.$$

[1]. Since we are told that  $v_2$  is non zero, this tells us the expected value for  $\hat{T}$  is larger than  $T$  and therefore the estimator is biased with  $\text{Bias}(\hat{T}) = \frac{\pi m}{8k} \frac{v_2}{n}$  [1]. Throughout the above, marks are deducted for missing major steps in the reasoning, it is not enough to just say that the expectation value of  $T$  is larger than  $T$  or that it is biased or similar. [8 marks]

- (d) A second estimator,  $\hat{T}'$  can be defined based on the second algebraic moment  $\mu_2 = \frac{3kT}{m}$  and the second algebraic sample moment for the data,  $m_2 = \frac{1}{n} \sum v_i^2$ . By equating  $\mu_2$  and  $m_2$ , find the expression for  $\hat{T}'$  and show that this estimator

[This question continues on the next page ...]

is unbiased.

**ANSWER:** UNSEEN: In this question, we set  $\mu_2 = m_2$  [1] so that,

$$\frac{3k\hat{T}'}{m} = \frac{1}{n} \sum_{i=1}^n v_i^2 \implies \hat{T}' = \frac{m}{3kn} \sum_{i=1}^n v_i^2.$$

[1] To calculate the bias of this estimator, we calculate the expectation value of  $\hat{T}'$ ,

$$E[\hat{T}'] = E\left[\frac{m}{3kn} \sum_{i=1}^n v_i^2\right] = \frac{m}{3kn} \sum_{i=1}^n E[v_i^2].$$

[1] We are given the second algebraic moment  $\mu_2$  in the question so one just has to remember that the  $n$ -th algebraic moment of a random variable  $X$  is given by  $\mu_n = E[X^n]$  so  $E[v_i^2] = \mu_2 = \frac{3kT}{m}$  [1]. Plugging this in we get,

$$E[\hat{T}'] = \frac{m}{3kn} \sum_{i=1}^n \frac{3kT}{m} = \frac{m}{3kn} \frac{3kT}{m} \cdot n = T.$$

[1] Since the expectation  $E[\hat{T}']$  is equal to  $T$ , the estimator  $\hat{T}'$  is an unbiased estimator [1]. [6 marks]

- (iii) Name one other property, aside from being unbiased, of a *good* estimator and describe what that property means. You can use equations and/or diagrams in your answer.

**ANSWER:** SEEN: Any of the following examples from the lectures could be described;

- Consistency: An estimator  $\hat{\theta}$  is a *consistent* estimator if in the limit of large sample sizes, the estimator converges in probability to the true value  $\theta$ . This can be written as for all  $\epsilon > 0$ ,  $\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$  or some other expression for the limit.
- Efficiency (or minimum variance): An estimator  $\hat{\theta}$  is an *efficient* estimator if its variance reaches the theoretical minimum set by the Cramer-Rao bound  $V(\hat{\theta}) = \frac{1}{I(\theta)}$  where  $I(\theta) = E\left[\left(\frac{d(\ln f(X;\theta))}{d\theta}\right)^2\right]$  is the Fisher information. Answers might also be given defining the Fisher information using the second derivative instead. Full marks given if no expression for the Fisher information is given but some mention of the theoretical minimum bound is required.
- Minimum mean squared error: Not covered in lectures, but one could say that a good estimator is one that minimises the mean squared error defined by  $MSE(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + V(\hat{\theta})$  which is a combination of the bias and variance. The answer should define each of these terms.
- Vanishing bias and/or variance: Again, not seen in lectures but a good estimator can be one for which the limit of the bias and or variance (or the *MSE*) are zero in the limit of large data sample sizes. Again, the terms bias, variance etc must be defined.

[This question continues on the next page ...]

- Invariance: A good estimator might be one that is invariant with respect to transformations of the random variable, eg  $X \rightarrow X^2 \implies \hat{\theta}(X) = \hat{\theta}(X^2)$  and/or transformations of the parameterisation, eg for some function  $\alpha = g(\theta)$ , the estimator  $\hat{\alpha} = g(\hat{\theta})$ .

Other properties may be known but generic properties like “easy to calculate” receive no marks. Talking about error rates or power also receives no marks as that is a confusion with hypothesis testing. One mark is awarded for the name of the property [1] and two marks for a clear explanation of what it means including any equations or diagrams used [2].

[3 marks]

[Total 25 marks]

2. Hypothesis testing is the frequentist process of rejecting (or accepting) a scientific hypothesis given some observation in experimental data. Formally two hypotheses are defined,  $H_0$  := the null hypothesis, and  $H_1$  := the alternate hypothesis.

- (i) (a) State the definition of Type-I and Type-II errors in terms of the two hypotheses  $H_0$  and  $H_1$ .

**ANSWER:** SEEN in Lectures:

Answers can be given in the form of words or equations.

The Type-I error is the probability to reject the null hypothesis when the null hypothesis is true and the Type-II error is the probability to reject the alternate hypothesis (or accept the null hypothesis), when the alternate hypothesis is true.

$$\text{type - I error} = \alpha = P(X \in w | H_0)$$

[1]

$$\text{type - II error} = \beta = P(X \in W - w | H_1)$$

[1] where  $X \in W$  is some test statistic and  $w$  is the critical region [1].

Answers that just reference integrals are ok provided that the region of integration is specified and not restricted to the 1D lower or upper bound integral case used in lectures for  $p$ -values. For example, it's ok to write,

$$\text{type - I error} = \int_w f(X|H_0) dX$$

$$\text{type - II error} = \int_{W-w} f(X|H_1) dX$$

[3 marks]

- (b) SEEN in Lectures: A testing apparatus is designed to detect for a faulty piece of lab equipment. The test will report 'N' if the equipment is faulty and 'T' if the equipment is fine. The designer claims that the probability that the test reads 'N' when equipment is faulty is 75%. You test out a piece of equipment in your lab with the apparatus and get a reading 'N'. Which of the following statements is true given only the information you have in the question.

- A The probability that the equipment is faulty is 75%.
- B If the equipment is not faulty, the probability that the test yields 'N' is 25%.
- C The probability that the test would have read 'T' if the equipment was faulty is 25%

**ANSWER:** Only the third statement - C - can be made given the information in the question [2]. No marks are awarded if either other statement is also mentioned as true. [2 marks]

- (ii) An experiment is designed to detect decays of muon particles after being produced in an accelerator. A detector is setup to record the decay times,  $t$ , after the muon

[This question continues on the next page ...]

is produced at  $t_0 = 0$ . The experiment runs until a maximum time  $T$  such that no decay times  $t > T$  are recorded. The probability to observe a decay at a time  $t$  after is given by the exponential probability distribution,

$$f(t; \tau) = N \exp\left(-\frac{t}{\tau}\right)$$

where  $\tau$  is the *lifetime* parameter.

- (a) Derive the normalisation constant  $N$  in terms of  $\tau$  and  $T$ .

**ANSWER:** SEEN in Lectures: Using the property that the probability density should be normalised to 1 [1],

$$1 = \int_0^T N \exp\left(-\frac{t}{\tau}\right) dt = N \left[ -\tau \exp\left(-\frac{t}{\tau}\right) \right]_{t=0}^{t=T} = \tau N \left( 1 - \exp\left(-\frac{T}{\tau}\right) \right)$$

[1] hence,

$$N = \frac{1}{\tau \left( 1 - \exp\left(-\frac{T}{\tau}\right) \right)}$$

[1]

[3 marks]

- (b) The figure below (Figure 1) shows a distribution in data (observed values of the decay time  $t$ ) collected by the experiment, and the exponential probability distribution.

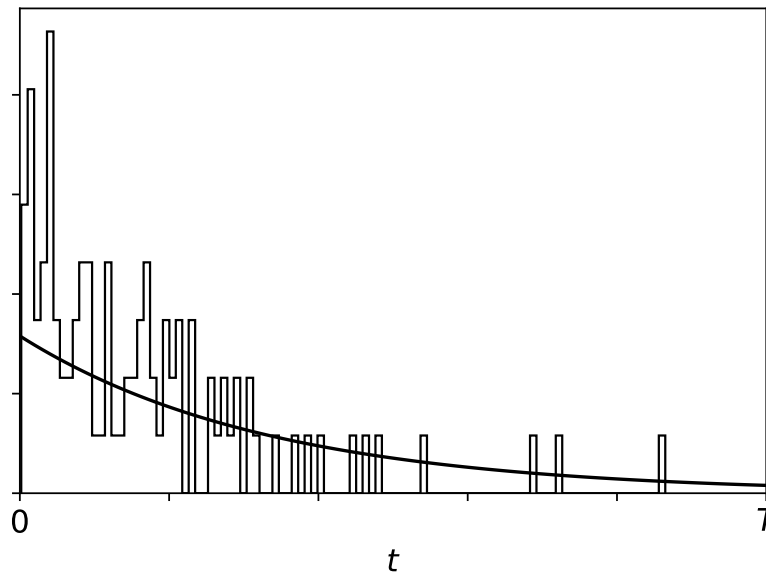


Figure 1

Describe how you would calculate the cumulative empirical distribution function  $F_{emp}(t)$  for the data.



**ANSWER:** The empirical distribution function for the observed data is given by the following expression,

$$F_{emp}(t) = \frac{\text{number of observations smaller than } t}{\text{total number of observations}}.$$

[2]. Marks are awarded for each of the numerator and denominator. An expression using the same bins as shown in the figure is also acceptable.

[2 marks]

(c) The Kolmogorov-Smirnov test-statistic is given by,

$$D = \max \left\{ |F_{emp}(t) - F(t)| \right\},$$

where  $F(t)$  is the cumulative distribution of the exponential function in this case. Derive  $F(t)$ .

**ANSWER:** UNSEEN: The answer is given by calculating the cumulative distribution function of the exponential probability density in terms of  $\tau$  and  $T$ ,

$$F(t; \tau) = \frac{1}{\tau \left(1 - \exp\left(-\frac{T}{\tau}\right)\right)} \int_0^t \exp\left(-\frac{t'}{\tau}\right) dt' = \frac{\tau \left(1 - \exp\left(-\frac{t}{\tau}\right)\right)}{\tau \left(1 - \exp\left(-\frac{T}{\tau}\right)\right)} = \frac{\left(1 - \exp\left(-\frac{t}{\tau}\right)\right)}{\left(1 - \exp\left(-\frac{T}{\tau}\right)\right)}$$

[3]. One mark for writing down the integral and two for reaching the correct expression. If the normalisation factor from part (a) is wrong, one mark for obtaining the right expression using the wrong normalisation is given.

[3 marks]

(d) A test is performed to see if the exponential is a good description for the data shown. The figure below (Figure 2) shows the distribution of  $D$  assuming the exponential shown is the true distribution ( $f(D|H_0)$ ) and the observed value of the test statistic  $D_{obs}$ . The figure also shows a value  $D_{95}$  for which 95% of the distribution is to the left of the line. Describe how the information in the figure could be used to test if the hypothesis that the exponential is a good description of the data is true or not. Based on this, do you think the exponential is a good description of the data or not?

**ANSWER:** UNSEEN in this specific form: The answer will depend on a threshold for the type-I error chosen by the experimenter. One answer could be that if the type-I error is chosen to be 0.05, since the observed value is above the threshold line, we reject  $H_0$  at this type-I error rate. The answer should explain that the critical region should be in the tail of the distribution since  $D$  will be larger when the data is more discrepant compared to the probability distribution. [2]. Only one mark is awarded for not explaining the reason for the choice of critical region (or why a tail integral is used).

[2 marks]

(iii) With the experimental data from part (ii), it should be possible to measure the parameter  $\tau$ . Assume for the rest of this question that the times observed are much smaller than the maximum time  $T$  or equally that  $T \rightarrow +\infty$ , and that the observed times are independent and known precisely.

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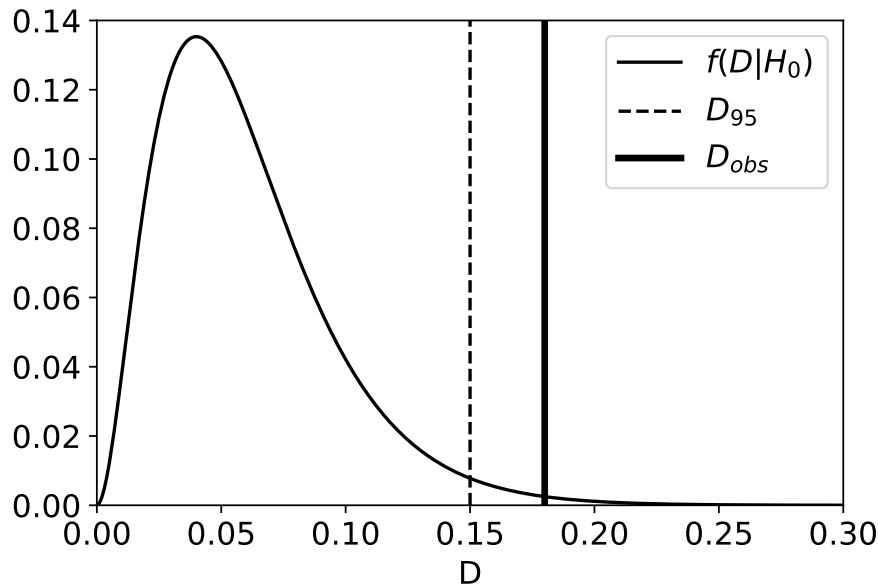


Figure 2

- (a) Write down the likelihood function for the set of  $n$  observations  $\{t_1, t_2, \dots, t_n\}$  assuming the same exponential probability distribution from part (ii).

**ANSWER:** Since  $\tau \rightarrow +\infty$ , the probability density simplifies to,

$$f(t; \tau) = \frac{1}{\tau} \exp\left(-\frac{t}{\tau}\right).$$

[1] The likelihood function is given by,

$$L(\tau) = \prod_{i=1}^n f(t_i; \tau) = \prod_{i=1}^n \frac{1}{\tau} \exp\left(-\frac{t_i}{\tau}\right)$$

[1]

[2 marks]

- (b) Derive the maximum likelihood estimate of the parameter  $\hat{\tau}$  and express the answer in terms of a sample moment of the dataset.

**ANSWER:** The maximum likelihood estimate  $\hat{\tau}$  is the value of  $\tau$  for which the likelihood is maximised. This is the same as minimising the negative of the natural log of the likelihood.

$$q = -\ln L(\tau) = \sum_{i=1}^n \ln(\tau) + \sum_{i=1}^n \frac{t_i}{\tau} = n \ln(\tau) + \sum_{i=1}^n \frac{t_i}{\tau}$$

[1] now, to find the minimum, we take the derivative of  $q$  with respect to  $\tau$  and set to zero.

$$0 = \frac{dq}{d\tau} \Big|_{\tau=\hat{\tau}} = \frac{n}{\hat{\tau}} - \frac{1}{\hat{\tau}^2} \sum_{i=1}^n t_i.$$

[This question continues on the next page ...]

[1] Multiplying through by  $\frac{\hat{\tau}}{n}$ , we have,

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n t_i = \bar{\tau},$$

which is the maximum likelihood estimator for  $\tau$  and is the first algebraic sample moment or sample mean of the times [1]. [3 marks]

- (c) Explain how you would construct a 68% confidence interval for the parameter  $\tau$ , using the Neyman construction. You should clearly state all of the steps required and you can write your answer in terms of pseudo-code if you wish.

**ANSWER:** The answer should contain the following set of steps:

1. Choose a test statistic  $t$  for the data. For example, this could be the based on log likelihood ratio  $t = -2 \ln \left( \frac{L(\tau)}{L(\hat{\tau})} \right)$  [1]
2. Calculate the value of the test statistic for the data  $t_{obs}$ , which could depend on  $\tau$ . [1]
3. For a range of  $\tau$  values derive the distribution  $f(t; \tau)$  of the test-statistic assuming  $\tau$  is the true value. Choose a procedure to construct 68% of the distribution in  $t$  for each value of  $\tau$ . Any well motivated procedure is ok, for example the likelihood ratio ordering principle is ok, or if the test-statistic is the one based on the log likelihood ratio, largest values could be included first. [1]
4. For each value of  $\tau$  determine whether the observed value  $t_{obs}$  lies inside the 68% region or not. This could be based on calculating a p-value  $p = \int_{t_{obs}}^{\infty} f(t; \tau) dt$  and choosing values of  $\tau$  for which  $p > 1 - 0.68 = 0.32$ . [1]
5. Join the values chosen or construct the union of values to form an interval (at 68% confidence level) in  $\tau$ . [1]

Each step must be included somewhere though often the last two steps will be merged, which is fine. [5 marks]

[Total 25 marks]

3. Two sets of data,  $\mathbf{d}_1$  and  $\mathbf{d}_2$  are collected, and used to infer parameters  $\theta$  of a model. In a Bayesian analysis, where the prior state of knowledge before either set of data is collected is  $\pi(\theta)$ .

- (i) Find an expression for the final posterior  $p(\theta|\mathbf{d}_1, \mathbf{d}_2)$  computed by a two-step process, first updating the original prior  $\pi(\theta)$  to a posterior given  $\mathbf{d}_1$ ,  $p(\theta|\mathbf{d}_1)$ , and using this as a prior for the analysis of  $\mathbf{d}_2$ . and show that this gives the same result as updating our state of knowledge from  $\pi(\theta)$  with the information from the combined dataset  $\mathbf{d}_1, \mathbf{d}_2$ . i.e. show that the two-step analysis yields the expected

$$p(\theta|\mathbf{d}_2, \mathbf{d}_1) = \frac{p(\mathbf{d}_2, \mathbf{d}_1|\theta)\pi(\theta)}{p(\mathbf{d}_2, \mathbf{d}_1)}.$$

ANSWER: SEEN:

$$p(\theta|\mathbf{d}_1) = \frac{p(\mathbf{d}_1|\theta)p(\theta)}{p(\mathbf{d}_1)}.$$

[1]

Now we analyse the second data set. We have some extra information from the first experiment, so we should update the prior to include  $\mathbf{d}_1$ , and Bayes' theorem applied to the second dataset gives a posterior

$$p(\theta|\mathbf{d}_2, \mathbf{d}_1) = \frac{p(\mathbf{d}_2|\theta, \mathbf{d}_1)p(\theta|\mathbf{d}_1)}{p(\mathbf{d}_2|\mathbf{d}_1)}.$$

[2]

We now notice that the new prior in this expression is just the old posterior probability, i.e. we have updated our prior state of knowledge from the original prior, instead using the posterior from the first dataset. We can also use the rules of probability to write the new likelihood as

$$p(\mathbf{d}_2|\mathbf{d}_1, \theta) = \frac{p(\mathbf{d}_2, \mathbf{d}_1|\theta)}{p(\mathbf{d}_1|\theta)}.$$

[2]

Substituting in the old posterior probability and this expression for the new likelihood we find

$$p(\theta|\mathbf{d}_2, \mathbf{d}_1) = \frac{p(\mathbf{d}_2, \mathbf{d}_1|\theta)\pi(\theta)}{p(\mathbf{d}_2, \mathbf{d}_1)}.$$

This has the same form as the outcome from the initial experiment, but now with the new data incorporated, i.e. the result we would write down if we analysed the data together ( $d \rightarrow \{\mathbf{d}_1, \mathbf{d}_2\}$ ). So, analysing separately and updating the prior after the first dataset gives the same answer as analysing the combined dataset with the original prior. [4] [9 marks]

- (ii) A physics theory makes a prediction for the probability  $q$  that an experiment detects a particle. An experiment is performed  $N$  times, and  $M$  independent detections are made.

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- (a) Argue that the probability of obtaining this outcome is the binomial distribution,

$$p(M|N, q) = \binom{N}{M} q^M (1-q)^{N-M}.$$

**ANSWER:** SEEN from Part I: Each detection has probability  $q$  and each non-detection  $1-q$ . Since they are independent trials, we multiply the probabilities. However, there are multiple orderings that give the same number of detections, given by the combinatorial prefactor [2].

- (b) A conjugate prior for the binomial distribution is the beta distribution,

$$\pi(q) = A(\alpha, \beta) q^{\alpha-1} (1-q)^{\beta-1}$$

where  $\alpha$  and  $\beta$  are hyperparameters of the prior, and  $A$  serves to normalise the probability distribution. What is a conjugate prior?

**ANSWER:** SEEN: A conjugate prior is one which, when combined with a likelihood function of a particular form, yields a posterior which has the same functional form as the prior, but with different parameters. [3]

- (c) Describe qualitatively what prior knowledge is reflected in values  $(\alpha, \beta) = (1, 1), (2, 1/4), (2, 2)$ .

**ANSWER:** UNSEEN:  $(1, 1)$  is uniform.  $(2, 1/4)$  puts high probability near  $q = 1$ .  $(2, 2)$  is symmetric, peaking in the centre. [2]

[7 marks]

- (iii) (a) Assuming a beta distribution prior, find an expression for the posterior for the experiment above.

**ANSWER:** UNSEEN:

$$\begin{aligned} p(q|M, N) &\propto p(M|q, N) \pi(q|N) \propto p(M|q, N) \pi(q) \\ &\propto A(\alpha, \beta) q^{\alpha-1} (1-q)^{\beta-1} \binom{N}{M} q^M (1-q)^{N-M} \\ &\propto q^{M+\alpha-1} (1-q)^{N-M+\beta-1}. \end{aligned} \quad (1)$$

[3]

- (b) Show that if two sets of trials are undertaken, yielding  $M_1$  detections from  $N_1$  trials, and then a further  $M_2$  detections from  $N_2$  trials, find explicit expressions for the final posterior (up to a normalising constant that you do not need to consider) and show that they are the same whether the sets are analysed in sequence, or the data combined first and analysed only once. Use the beta prior as before.

**ANSWER:** UNSEEN: Using the posterior from the first set, given by the previous expression but with  $M \rightarrow M_1$  and  $N \rightarrow N_1$ , as the prior, the final posterior

is

$$\begin{aligned}
 p(q|M_1, M_2, N_1, N_2) &\propto p(M_2|q, N_2)\pi(q|N_2) \propto p(M_2|q, N_2)p(q|M_1, N_1) \\
 &\propto q^{\alpha-1}(1-q)^{\beta-1} \binom{N_2}{M_2} q^{M_2}(1-q)^{N_2-M_2} q^{M_2}(1-q)^{N_2-M_2} \\
 &\propto q^{M+\alpha-1}(1-q)^{N-M+\beta-1}, \quad (2)
 \end{aligned}$$

where  $M = M_1 + M_2$  and  $N = N_1 + N_2$ , as required. [3]

[6 marks]

- (iv) (a) Find the MAP (maximum a posteriori) value of  $q$ , for given  $\alpha$  and  $\beta$ . If  $\alpha + \beta = 2$ , comment on how it depends on  $\alpha$ , as well as on the data. You may wish to simplify the algebra by defining  $C = M + \alpha - 1$  and  $D = N - M + \beta - 1$ .

**ANSWER:** UNSEEN: The answer is obtained by maximising the posterior w.r.t.  $q$ :

$$\frac{d \ln p}{dq} = \frac{C}{q} - \frac{D}{1-q}$$

where  $C = M + \alpha - 1$  and  $D = N - M + \beta - 1$ , and setting this to zero gives

$$q_{\text{MAP}} = \frac{C}{C+D} = \frac{M + \alpha - 1}{N + \alpha + \beta - 2}.$$

For a uniform prior, this is  $M/N$ , as one might expect. If  $\alpha < 1$ , then the prior favours low values of  $q$ , and the posterior peaks below  $M/N$ . If  $\alpha > 1$ , then  $\beta < 1$ , and the prior favours values of  $q$  near 1, so the MAP value exceeds the naive value of  $M/N$  from the data. [4]

- (b) By expanding the log(posterior) around the peak value, show that the uncertainty on  $q$  may be written as

$$\sigma_q^{-2} = \frac{(N + \alpha + \beta - 2)^3}{(N - M + \beta - 1)(M + \alpha - 1)}.$$

**ANSWER:** UNSEEN: We approximate the inverse variance by minus the second derivative of  $\ln(p)$  at the peak.

$$\sigma_q^{-2} \simeq -\frac{d^2 \ln(p)}{dq^2} = \frac{C}{q^2} + \frac{D}{(1-q)^2} = \frac{C}{C^2}(C+D)^2 + \frac{D}{D^2}(C+D)^2 = \frac{(C+D)^3}{CD}$$

Hence result. [2]

[6 marks]

[Total 28 marks]

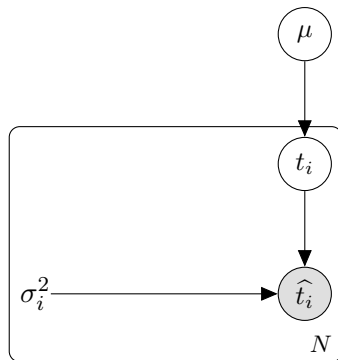


Figure 1: DAG.

4. The decay rate  $\mu > 0$  for a radioactive process (whose signal is proportional to  $\exp(-\mu t)$  for  $t \geq 0$ ) is to be inferred from independent measurements  $\hat{t}_i$  ( $i = 1, \dots, N$ ) of decay times. The measurements are known to be imprecise, with uncorrelated gaussian errors of known variances  $\sigma_i^2$ .

- (i) (a) What are latent variables and which ones occur in this problem?

**ANSWER: SEEN:** A latent variable is part of the data model, but is not directly measured. [2]. **UNSEEN:** The set of true decay times  $t_i$ . [2]

- (b) If  $t$  is the time of a decay, write down a properly normalised expression for the sampling distribution of  $t$ .

**ANSWER: UNSEEN:**  $p(t|\mu) = \mu \exp(-\mu t)$ . [1]

[5 marks]

- (ii) (a) Draw a directed acyclic graph (DAG) that represents the process, identifying quantities that are drawn from probability distributions with open circles, known quantities with a dot, deterministic quantities in a diamond, and quantities drawn from the same distribution should be surrounded with a rectangular 'plate', identifying the number of repetitions. Measured data should be in shaded circles. Include any latent variables identified in the previous part.

**ANSWER: UNSEEN:** See the diagram. Note that there should be a dot next to  $\sigma_i^2$ , but the tikz software doesn't draw one.  $t_i$  are the latent variables.

[6 marks]

- (iii) (a) For a given prior  $\pi(\mu)$ , write down an expression for the posterior for  $\mu$  using a Bayesian hierarchical model based on the DAG. Make use of the rules of probability and marginalise over the latent variables, to write your answer in terms of a product of integrals.

[This question continues on the next page ...]

**ANSWER: UNSEEN:**

$$\begin{aligned}
 p(\mu|\{\hat{t}_i\}) &\propto p(\{t_i\}|\mu)\pi(\mu) \\
 &\propto \int p(\{\hat{t}_i\}, \{t_i\}|\mu) d\{t_i\} \pi(\mu) && \text{Marginalise over latent variables} \\
 &\propto \int p(\{\hat{t}_i\}|\{t_i\}, \mu) p(\{t_i\}|\mu) d\{t_i\} \pi(\mu) \\
 &\propto \prod_{i=1}^N \left[ \int p(\hat{t}_i|t_i, \mu) p(t_i|\mu) dt_i \right] \pi(\mu) && \text{Independent} \\
 &\propto \prod_{i=1}^N \left[ \int p(\hat{t}_i|t_i) p(t_i|\mu) dt_i \right] \pi(\mu) && \text{No } \mu \text{ dependence in first term} \\
 &\propto \prod_{i=1}^N \left[ \int \exp\left[-\frac{1}{2\sigma_i^2} (\hat{t}_i - t_i)^2\right] \mu \exp(-\mu t_i) dt_i \right] \pi(\mu)
 \end{aligned}$$

[8 marks]

(iv) Prove that the posterior is given by

$$p(\mu|\{\hat{t}_i\}) \propto \mu^{N-1} \prod_{i=1}^N \exp\left(-\mu \hat{t}_i + \frac{\mu^2 \sigma_i^2}{2}\right) \left[1 + \operatorname{erf}\left(\frac{\hat{t}_i}{\sqrt{2}\sigma_i} - \frac{\mu\sigma_i}{\sqrt{2}}\right)\right]$$

for a particular choice of prior, and justify that prior. Use the gaussian integral given at the end of the question and explicitly write down expressions for  $a$ ,  $b$ , and  $c$ .

**ANSWER: UNSEEN:** Expand out the quadratic in the previous part, and use the supplied integral.

$$\begin{aligned}
 a &= \frac{1}{2\sigma_i^2} \\
 b &= \frac{\hat{t}_i}{\sigma_i^2} - \mu \\
 c &= \frac{\hat{t}_i^2}{2\sigma_i^2}
 \end{aligned}$$

We have assumed a Jeffreys prior  $\pi(\mu) \propto 1/\mu$ , since  $\mu > 0$  is a scaling parameter.

[6 marks]

You may assume without proof the result

$$\int_0^\infty \exp(-ax^2 + bx + c) dx = \sqrt{\frac{\pi}{4a}} \exp\left(\frac{b^2}{4a} + c\right) \left[1 + \operatorname{erf}\left(\frac{b}{2\sqrt{a}}\right)\right].$$

[Total 25 marks]